

LECTURE 9 (SEPTEMBER 30TH)

Definition. (14.2) Let $f(t) \in K[t]$. A splitting field for $f(t)$ is a field extension F/K such that $f(t)$ factors into linear terms

$$c \cdot \prod_{i=1}^n (t - \alpha_i)$$

with $F = K(\alpha_1, \alpha_2, \dots, \alpha_n)$.

Definition. (13.19) A field F is algebraically closed if every nonconstant polynomial $f(t) \in F[t]$ has a zero in F .

Remark. If $K = \mathbf{Q}$, the existence of a the splitting field of $f(t)$ is immediate: if $f(t) = c \cdot \prod_{i=1}^n (t - \alpha_i)$ in $\mathbf{C}[t]$, then

$$F = K(\alpha_1, \dots, \alpha_n) \subset \mathbf{C}$$

Theorem. (14.6) There exists a splitting field of $f(t) \in K[t]$.

Proof. Let simply lay out the idea. For any $f(t) \in K[t]$, $f(t) = p_1(t)f_1(t)$ where $p_1(t)$ is irreducible. By Kronecker's theorem, we can set $K_1 = K(\alpha_1)/K$, where $p_1(\alpha_1) = f(\alpha_1) = 0$. Then,

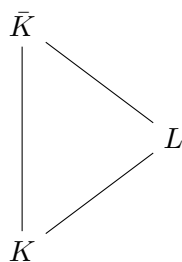
$$p_1(t) = p_2(t)f_3(t)$$

and we can set $K_2 = K_1(\alpha_2)$ where $p_2(\alpha_2) = 0$. Thus,

$$f(t) = (t - \alpha_1)(t - \alpha_2)g(t) \in K_2[t]$$

and so on. □

Theorem. (14.3) Let K be a field. Then there exists an algebraic extension $\bar{K}/K$ such that $\bar{K}$ is algebraically closed.



Remark. Such an extension is unique: if F and L are two such extensions of K , then there exists an isomorphism $\sigma : F \rightarrow L$ such that $\sigma|_K = \text{id}_K$ (that is, $\sigma(a) = a$ for all $a \in K$).

$$\begin{array}{ccc}
 F & \xrightarrow{\sigma} & L \\
 \uparrow & & \uparrow \\
 K & \xrightarrow{1} & K
 \end{array}$$

Definition. (14.4) We call $\bar{K}$ the algebraic closure of K .

LECTURE 10 (OCTOBER 2ND)

Remark. Last class we talked about algebraically closed fields. This class we will be talking about isomorphism extensions theorem.

Definition. (13.19) A field K is algebraically closed if for every algebraic extension F of K , $F = K$, or equivalently, every nonconstant polynomial $f(t) \in K[t]$ splits into linear factors.

Definition. (14.4) $\bar{K}$ is an algebraic closure of K if

- (i) $\bar{K}/K$ is algebraic
- (ii) $\bar{K}$ is algebraically closed

Theorem. (Isomorphic extension) Let $\sigma : K_1 \rightarrow K_2$ be an isomorphism of fields. Let F/K_1 be an algebraic extension. Then there exists a field embedding $\bar{\sigma} : F \rightarrow \bar{K}_2$ extended σ (that is, $\bar{\sigma}|_{K_1} = \sigma$).

$$\begin{array}{ccc}
 F & \xrightarrow{\bar{\sigma}} & \bar{K}_2 \\
 \uparrow & & \uparrow \\
 K_1 & \xrightarrow{\sigma} & K_2
 \end{array}$$

Proposition. (14.1) Let F/K be a field extension and let $p(x) \in K[x]$ be irreducible. Then there is a 1-1 correspondence between the sets of zeros of $p(t)$ in F and the set of K -embeddings $K[t]/(p(t)) \rightarrow F$ which extends the inclusion $K \hookrightarrow F$.

Proof. Suppose that $\alpha \in F$ is a zero of $p(t)$. Consider the evaluation homomorphism $\text{ev}_\alpha : K[t] \rightarrow F$. Then, $\ker(\text{ev}_\alpha) = (p(t))$. Conversely, if $\sigma : K[t]/(p(t)) \rightarrow F$ is a K -embedding, then $\sigma(\bar{t}) \in F$ is a zero of $p(t)$ in F .

$$\begin{array}{ccc}
K[t]/(p(t)) & \longrightarrow & F \\
\uparrow & & \uparrow \\
K & \xrightarrow{\mathbf{1}} & K
\end{array}$$

□

Remark. We now write $\sigma : R_1 \rightarrow R_2$ for a ring homomorphism. If $f(t) \in R_1[t]$ is a polynomial, we let $f_\sigma(t)$ denote the image of $f(t)$ under the induced homomorphism $R_1[t] \rightarrow R_2[t]$. Concretely, for $f(t) = \sum_{i=0}^n a_i t^i$ we have $f_\sigma(t) = \sum_{i=0}^n \sigma(a_i) t^i$.

Proposition. Let $\sigma : K_1 \rightarrow K_2$ be an isomorphism of fields. Let F_1/K_1 and F_2/K_2 be field extensions. Let $p(t)$ be an irreducible polynomial in $K_1[t]$. Suppose that $\alpha \in F_1$ is a zero of $p(t)$ and that $\beta \in F_2$ is a zero of $p_\sigma(t)$. Then there exists a unique isomorphism $\bar{\sigma} : K_1(\alpha) \rightarrow K_2(\beta)$ such that $\bar{\sigma}|_{K_1} = \sigma$ and that $\bar{\sigma}(\alpha) = \beta$.

$$\begin{array}{ccc}
F_1 & & F_2 \\
\uparrow & & \uparrow \\
K_1(\alpha) & \xrightarrow{\bar{\sigma}} & K_2(\beta) \\
\uparrow & & \uparrow \\
K_1 & \xrightarrow{\sigma} & K_2
\end{array}$$

with irreducible

$$\begin{cases} p(t) \in K_1[t] \\ p(\alpha) = 0 \end{cases} \quad \& \quad \begin{cases} p_\sigma(t) \in K_2[t] \\ p_\sigma(\beta) = 0 \end{cases}$$

Proof. (I) Define $\bar{\mathbf{1}} : K(\alpha) \rightarrow K(\beta)$ by sending $f(\alpha)/g(\alpha) \in K(\alpha)$ to $f(\beta)/g(\beta) \in K(\beta)$.

(II) Define the function $\bar{\mathbf{1}}$ through

$$\begin{array}{ccc}
K[t]/(p(t)) & & K[t]/(p(t)) \\
\uparrow & & \uparrow \\
K(\alpha) & \xrightarrow{\bar{1}} & K(\beta) \\
\uparrow & & \uparrow \\
K & \xrightarrow{1} & K
\end{array}
\quad \begin{array}{c} \\ \\ \circlearrowleft \\ \\ \end{array}$$

Keeping this in mind, we write

$$\mathrm{ev}_\beta \circ (\mathrm{ev}_\alpha)^{-1}$$

□